

Student Name: _____

Date: _____

Class / Set: _____

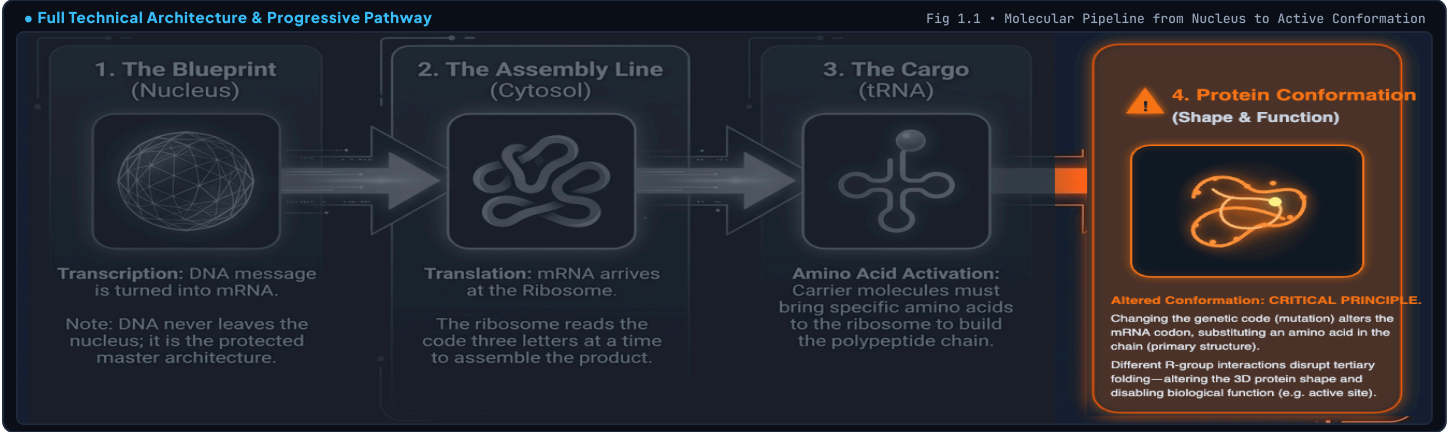
Target Grade: _____

RULE

THE CENTRAL DOGMA FRAMEWORK: STRUCTURE DETERMINES FUNCTION

Level 3 Essential Standard

To operate at a Level 3 standard, we must understand **molecular causality**: altering the DNA triplet code changes the amino acid sequence, directly disrupting tertiary folding, active site geometry, and biological function.



• ALTERED CONFORMATION: CRITICAL PRINCIPLE

Why Structure Strictly Dictates Function

Changing the genetic code (mutation) alters the **mRNA codon**, substituting a different amino acid into the polypeptide chain (**primary structure**). Different R-group chemical properties disrupt **tertiary folding** (hydrogen bonds, ionic bonds, disulfide bridges)—altering the 3D protein shape and disabling biological function (e.g. deforming the enzyme active site).

• The Level 3 Molecular Causality Chain (Memorise for Full Marks)

Sequential Step-by-Step Mechanism

STEP 01

DNA Mutation

Base substitution alters triplet code sequence in the gene.

STEP 02

Altered mRNA

Transcription produces mRNA with an altered codon.

STEP 03

Different Amino Acid

tRNA delivers a substituted amino acid during translation.

STEP 04

R-Group Disruption

New R-group cannot form original ionic / hydrogen / disulfide bonds.

STEP 05

Altered Tertiary Shape

3D conformation folds abnormally, changing active site geometry.

STEP 06

Loss of Function

Substrate can no longer bind; zero enzyme-substrate complexes form!

• Disulfide Bridges

Covalent bonds between sulfur atoms in two cysteine R-groups. Strongest tertiary bond; stabilizes conformation.

• Ionic Bonds

Electrostatic attractions between oppositely charged R-groups (-NH₃⁺ and -COO⁻). Easily broken by pH changes.

• Hydrogen Bonds

Numerous weak attractions between polar R-groups (e.g. -OH and -NH). Individually weak, collectively vital; heat sensitive.

• Hydrophobic Forces

Non-polar R-groups fold inwards away from aqueous cytoplasm, driving spontaneous 3D compact globular folding.

• Level 3 Exam Application Task

6 Marks • High Yield Exam Question

Question: A patient carries a missense mutation involving a single base substitution in the gene encoding an essential metabolic enzyme. Explain step-by-step why this mutation causes the enzyme to lose its catalytic activity.

1. Effect on transcription:

(DNA triplet → mRNA codon) [1 Mark]

2. Primary structure & tRNA:

(Substituted amino acid in chain) [1 Mark]

3. R-group chemical bonding:

(Disrupted ionic, hydrogen, or disulfide bonds) [1 Mark]

4. 3D Tertiary folding:

(Abnormal folding & altered active site cleft) [1 Mark]

5. Substrate complementarity:

(Substrate no longer fits active site geometry) [1 Mark]

6. Biological & catalytic outcome:

(Zero enzyme-substrate complexes & loss of function) [1 Mark]

• Level 3 Examiner Mark Scheme Criteria (Tick Once Addressed in Your Answer)

Self-Assessment Checklist

☐ Stated DNA mutation changes the transcribed mRNA codon.

☐ Named disrupted bonding (hydrogen, ionic, or disulfide bridges).

☐ Explained active site is no longer complementary to substrate.

☐ Stated different amino acid alters primary structure sequence.

☐ Stated tertiary (3D) structure / conformation is altered.

☐ Concluded no enzyme-substrate complexes form → loss of activity.

Examiner's Top Tip for Distinction (Level 3): Never simply write "the protein doesn't work". Always link the chain: *Primary structure → R-group bonds → 3D tertiary conformation → active site complementarity → inability to form enzyme-substrate complexes.*

Slide 8 • Build 1

1. The Blueprint (Nucleus)

1. The Blueprint: Transcription in the Nucleus

Focus: Nuclear protection & working transcript generation

Core Framework Breakdown & Technical Architecture Deep Dive

The Central Dogma Framework: Structure determines function.

To operate at a Level 3 standard, we must understand molecular causality: altering the DNA triplet code changes the amino acid sequence, directly disrupting tertiary folding, active site geometry, and function.

1. The Blueprint (Nucleus)



Transcription: DNA message is turned into mRNA.

Note: DNA never leaves the nucleus; it is the protected master architecture.

• Guided Lecture Notes (Record during teacher explanation):

1. Role of RNA Polymerase during transcription:

2. Why DNA is strictly restricted to the nucleus:

3. How mRNA exits through nuclear pores into cytosol:

Core Concept: DNA contains triplet codes transcribed into single-stranded messenger RNA (mRNA) by RNA polymerase. DNA never leaves the nucleus; it is the protected master architecture.

Slide 8 • Build 2

2. The Assembly Line (Cytosol)

2. The Assembly Line: Ribosomal Translation

Focus: Codon reading & peptide bond formation

Core Framework Breakdown & Technical Architecture Deep Dive

The Central Dogma Framework: Structure determines function.

To operate at a Level 3 standard, we must understand molecular causality: altering the DNA triplet code changes the amino acid sequence, directly disrupting tertiary folding, active site geometry, and function.

1. The Blueprint (Nucleus)



Transcription: DNA message is turned into mRNA.

Note: DNA never leaves the nucleus; it is the protected master architecture.

2. The Assembly Line (Cytosol)



Translation: mRNA arrives at the Ribosome.

The ribosome reads the code three letters at a time to assemble the product.

• Guided Lecture Notes (Record during teacher explanation):

1. Define codon and explain triplet reading frame:

2. Function of ribosome as the catalytic workbench:

3. Why non-overlapping translation prevents frame errors:

Core Concept: mRNA binds to the ribosome in the cytosol. The ribosome reads the code three letters at a time (codons) to assemble the product with strict reading frame fidelity.

AAQ Human Biology • Lesson 01: Welcome to Human Biology

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Slide 8 • Build 3

3. The Cargo (tRNA)

3. The Cargo: Amino Acid Activation & Delivery

Focus: Anticodon complementarity & primary sequence

• Guided Lecture Notes (Record during teacher explanation):

1. How amino acid activation ensures translation accuracy:

2. Matching between mRNA codon and tRNA anticodon:

3. The condensation reaction forming peptide bonds:

Core Framework Breakdown & Technical Architecture Deep Dive

The Central Dogma Framework: Structure determines function.

To operate at a Level 3 standard, we must understand molecular causality: altering the DNA triplet code changes the amino acid sequence, directly disrupting tertiary folding, active site geometry, and function.

1. The Blueprint (Nucleus)



Transcription: DNA message is turned into mRNA.

Note: DNA never leaves the nucleus; it is the protected master architecture.

2. The Assembly Line (Cytosol)



Translation: mRNA arrives at the Ribosome.

The ribosome reads the code three letters at a time to assemble the product.

3. The Cargo (tRNA)



Amino Acid Activation: Carrier molecules must bring specific amino acids to the ribosome to build the polypeptide chain.

Core Concept: Transfer RNA (tRNA) molecules are loaded with specific amino acids. The tRNA anticodon pairs complementarily with mRNA codons, building the polypeptide chain.

Slide 8 • Build 4

4. Protein Conformation & Mutation

4. Conformation: Shape Dictates Function

Focus: Molecular causality & mutation disruption

• Guided Lecture Notes (Record during teacher explanation):

1. Types of R-group bonds stabilizing 3D tertiary structure:

2. Why a point mutation alters R-group chemical interactions:

3. Why deformed active sites prevent substrate binding:

Core Framework Breakdown & Technical Architecture Deep Dive

The Central Dogma Framework: Structure determines function.

To operate at a Level 3 standard, we must understand molecular causality: altering the DNA triplet code changes the amino acid sequence, directly disrupting tertiary folding, active site geometry, and function.

1. The Blueprint (Nucleus)



Transcription: DNA message is turned into mRNA.

Note: DNA never leaves the nucleus; it is the protected master architecture.

2. The Assembly Line (Cytosol)



Translation: mRNA arrives at the Ribosome.

The ribosome reads the code three letters at a time to assemble the product.

3. The Cargo (tRNA)



Amino Acid Activation: Carrier molecules must bring specific amino acids to the ribosome to build the polypeptide chain.

4. Protein Conformation (Shape & Function)



Altered Conformation: CRITICAL PRINCIPLE.
Changing the genetic code (mutation) alters the mRNA codon, misidentifying an amino acid in the chain (primary structure).
Different R-group interactions disrupt tertiary folding—altering the 3D protein shape and disabling biological function (e.g. active site).

CRITICAL PRINCIPLE: Altering the DNA triplet changes the amino acid in the primary structure. Different R-group interactions disrupt tertiary folding—altering active site geometry and disabling function!